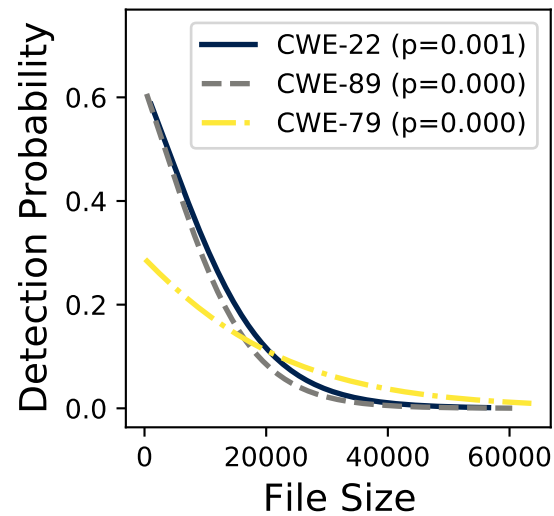
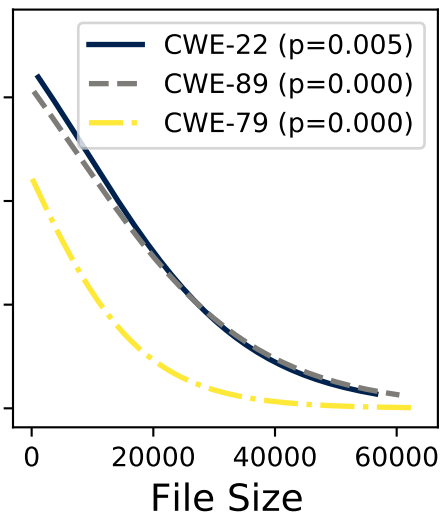


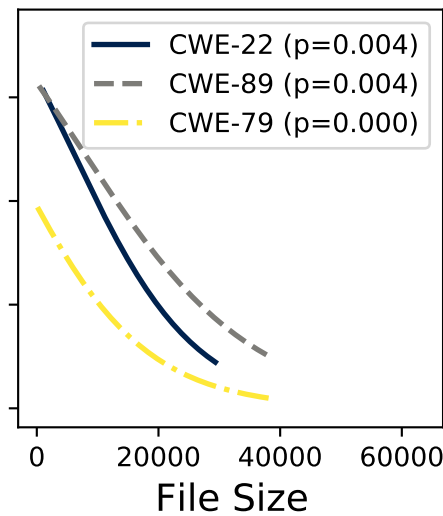
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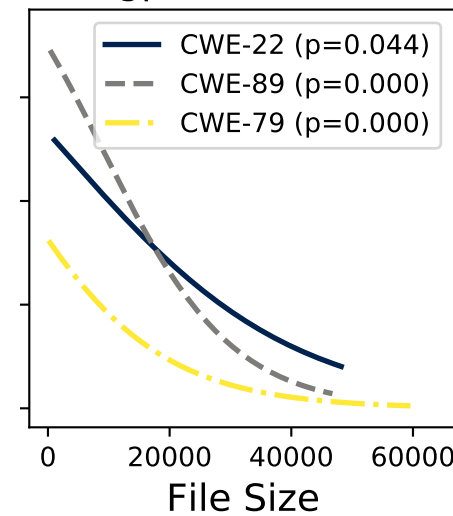
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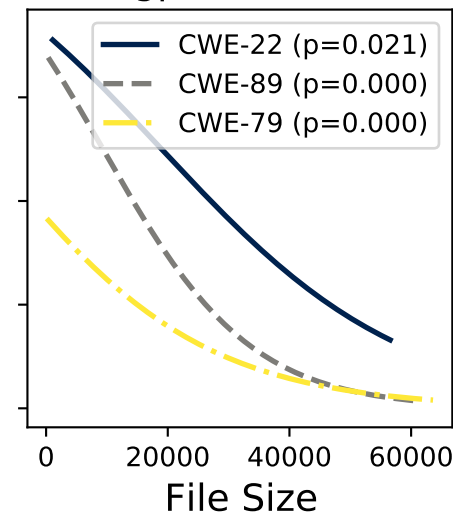
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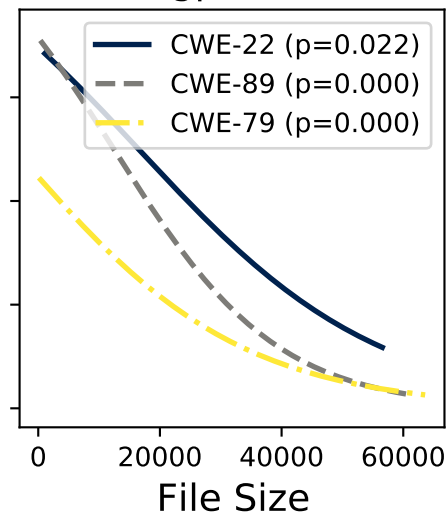
gpt-3.5-turbo



gpt-4-turbo



gpt-4o



Bug Position	CWE-22 (p=0.001)	CWE-89 (p=0.002)	CWE-79 (p=0.000)
0	0.62	0.52	0.28
10000	0.02	0.10	0.12
20000	0.01	0.02	0.05
30000	0.01	0.01	0.02
40000	0.01	0.01	0.01

Figure 10: Number of bugs found by CWE-22, CWE-89, and CWE-79 across different bug positions. The graph shows that the number of bugs found decreases as the bug position increases. CWE-22 (solid dark blue line) consistently finds the most bugs, followed by CWE-89 (dashed grey line), and CWE-79 (dash-dot yellow line) finds the fewest bugs across the range of positions shown.

Bug Position	CWE-22 (p=0.003)	CWE-89 (p=0.001)	CWE-79 (p=0.000)
0	~10	~8	~5
10000	~4	~3	~1.5
20000	~2	~1.5	~0.8
30000	~1	~0.8	~0.5
40000	~0.8	~0.6	~0.4

Figure 10: CWE-22, CWE-89, and CWE-79. This line graph illustrates the distribution of bugs found across different positions for three specific CWE categories. The x-axis, labeled 'Bug Position', ranges from 0 to 40,000. The y-axis represents the number of bugs found. Three data series are plotted: CWE-22 (dark blue solid line, $p=0.039$), CWE-89 (grey dashed line, $p=0.003$), and CWE-79 (yellow dash-dot line, $p=0.000$). All three series show a decreasing trend in the number of bugs found as the bug position increases. CWE-79 consistently shows the highest number of bugs found, followed by CWE-22 and then CWE-89.

The graph displays the number of bugs found by three specific CWEs across different bug positions. The x-axis represents the Bug Position (0 to 40,000), and the y-axis represents the number of bugs found (0 to 100). The legend indicates the following data series:

- CWE-22 ($p=0.017$): Solid dark blue line.
- CWE-89 ($p=0.002$): Dashed grey line.
- CWE-79 ($p=0.000$): Dash-dot yellow line.

Approximate data points extracted from the graph:

Bug Position	CWE-22 (p=0.017)	CWE-89 (p=0.002)	CWE-79 (p=0.000)
0	~75	~95	~55
10000	~45	~65	~35
20000	~25	~40	~20
30000	~15	~25	~15
40000	~10	~15	~10

Figure 10 is a line graph titled "Bug Position" on the x-axis, which ranges from 0 to 40,000. The y-axis is unlabeled but represents frequency. Three curves are plotted, each representing a different CWE category with its associated p-value:

- CWE-22 ($p=0.002$)**: Represented by a solid dark blue line. It starts at the highest frequency (approximately 45,000) at position 0 and decays rapidly, reaching near zero by position 30,000.
- CWE-89 ($p=0.000$)**: Represented by a dashed grey line. It starts at a frequency of approximately 35,000 at position 0 and decays rapidly, reaching near zero by position 30,000.
- CWE-79 ($p=0.000$)**: Represented by a dash-dot yellow line. It starts at a frequency of approximately 25,000 at position 0 and decays rapidly, reaching near zero by position 30,000.

All three curves show a similar trend of rapid decay, indicating that bugs for these categories are most frequently found at lower positions in the code.

Figure 10: The number of bugs found by CWE-22, CWE-89, and CWE-79 across different bug positions. The x-axis represents Bug Position (0 to 40,000) and the y-axis represents the number of bugs found. CWE-22 (solid dark blue line) starts highest at position 0 and decreases. CWE-89 (dashed grey line) starts second highest and decreases. CWE-79 (dash-dot yellow line) starts lowest and decreases most rapidly, crossing below CWE-22 around position 10,000.